

Fifty Supernovae and One Combination of Two Parameters

The Accelerating Universe as Statistical Inference, 1988–1999

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Abstract

In 1998 two teams reported that distant supernovae are fainter than a decelerating universe allows, and concluded that cosmic expansion is speeding up. The inference is a weighted nonlinear regression with two parameters of interest and a nuisance intercept. I run the Fisher-information calculation that prices it. The result reproduces the published error bar from the observing design alone, and shows that the supernovae measured one combination of the two density parameters and left the orthogonal combination nearly free. The famous numbers assume a flat universe. Ruling out a decelerating universe needs a prior the supernovae do not supply. I also set out the episode’s genuine out-of-sample test, which came six years later and had no free parameters, and I ask what the whole affair says about theory-free discovery. The derivations and the full data are in a longer companion version.

1 What Was Measured

In 1998 two teams looked at exploding stars billions of light-years away and found them fainter than they should have been. Fainter means further. Further, at a given redshift, means the universe has expanded more since the light left than anyone had allowed for. The expansion is speeding up, and something is pushing it. Nobody knows what.

The inference from those photometric measurements to that conclusion is a regression, and this note is about its statistical anatomy: what was identified, what was assumed, and how much of the famous result the data actually carry.

Start with the measurement. A *standard candle* is an object whose intrinsic brightness you know. Knowing it, you convert observed brightness into distance, since flux falls as the inverse square. Collect distances and redshifts for objects near and far and you have traced the expansion history of the universe.

*In 2011 my good friend Christopher A. Sims and I shared the Nobel Prize in economics. That good luck let us meet and become friends with winners of prizes in physics and other fields. Adam Riess had majored partly in economics as an undergraduate. He and Brian Schmidt have since taught me the nuts and bolts of the computer programs they used to climb the likelihood function that led to their discovery. This paper is one of five that read episodes in the history of physics and astronomy as exercises in statistical inference and machine learning. [Sargent \[2025\]](#) sets out the theme: what is now called artificial intelligence names practices that scientists have used for centuries. The other four treat Kepler and Newton, Einstein’s photoelectric equation, Pauli’s exclusion principle, and the exoplanet discovery of Mayor and Queloz.

Type Ia supernovae are thermonuclear explosions of white dwarfs in binary systems. Their peak luminosities are nearly, but not quite, uniform. The account current in the 1990s had the white dwarf accreting until it neared the Chandrasekhar limit, so that a near-universal exploding mass gave a near-universal brightness. That account no longer commands consensus. Sub-Chandrasekhar detonations and white-dwarf mergers are now taken seriously, and the progenitor population may be mixed. The uniformity is an empirical fact, not a derived one, and it was empirical in 1998 too.

Magnitudes are a logarithmic brightness scale on which five magnitudes is a factor of one hundred in flux, and larger means fainter. Raw Type Ia peak magnitudes scatter by about 0.4 magnitudes, a 20% distance error. The cosmological signal these teams were chasing is about 0.25 magnitudes. Before 1993 the noise was larger than the thing being measured.

Two groups went after it anyway. The Supernova Cosmology Project, led by Saul Perlmutter at Berkeley, began systematic high-redshift searches in the late 1980s and reported 42 supernovae at redshift $z = 0.18$ to 0.83 [Perlmutter et al., 1999]. The High-Z Supernova Search Team, assembled around Brian Schmidt with Adam Riess writing the analysis pipeline, reported 16 high-redshift events at $z = 0.16$ to 0.97 together with 34 nearby ones [Riess et al., 1998]. Both concluded that the expansion is accelerating. Perlmutter took half the 2011 Nobel Prize; Riess and Schmidt shared the other half.

2 The Regression

General relativity, plus the assumption that the universe looks the same in every direction and from every place, gives the expansion rate as

$$H^2(z) = H_0^2 [\Omega_M(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda], \quad (1)$$

where Ω_M is the density of matter, Ω_Λ the density of the cosmological constant, and Ω_k spatial curvature, all in units that make them sum to one. A *flat* universe has $\Omega_k = 0$. Integrating (1) along the light path gives the luminosity distance $d_L(z; \Omega_M, \Omega_\Lambda)$, and taking logarithms gives the *distance modulus* $\mu = 5 \log_{10}(d_L/\text{Mpc}) + 25$, which is what a supernova's apparent magnitude measures once its intrinsic magnitude is known.

The problem is well behaved for two reasons. The functional form is fixed, exactly, by physics established seventy years earlier. And the Hubble constant H_0 and the intrinsic magnitude M_B enter only through a single combination,

$$\mathcal{M} = M_B + 5 \log_{10} \left(\frac{c}{H_0 \text{ Mpc}} \right) + 25, \quad (2)$$

so the measurement equation is

$$m_{B,i} = \mathcal{M} + 5 \log_{10} \tilde{d}_L(z_i; \Omega_M, \Omega_\Lambda) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_i^2), \quad (3)$$

where $\tilde{d}_L$ is the dimensionless part of the distance that depends only on the density parameters.

$\mathcal{M}$ is a textbook nuisance parameter. It shifts the whole Hubble diagram vertically and leaves its shape alone. Neither H_0 nor M_B can be recovered from the diagram, because any change in one is offset exactly by a change in the other. Neither needs to be. Since $\mathcal{M}$ enters linearly it

can be concentrated out analytically at each candidate $(\Omega_M, \Omega_\Lambda)$, leaving a profile criterion in two parameters.

Millikan faced the same structure in 1916 measuring Planck’s constant. The contact voltage between his photoemitting metal and his collector shifted every reading by the same amount and left the slope he wanted untouched. The parallel is exact as far as it goes. Section 5 shows where it stops.

Under Gaussian errors, maximising the likelihood means minimising

$$\chi^2 = \sum_i \frac{[m_{B,i} - \mathcal{M} - 5 \log_{10} \tilde{d}_L(z_i; \Omega_M, \Omega_\Lambda)]^2}{\sigma_i^2}, \quad (4)$$

which is weighted least squares. Unlike Millikan’s line, $\tilde{d}_L$ has no closed form, so both groups minimised (4) on a grid, computing the integral numerically at each point.

The easy case, in the preceding note

The preceding note works through Millikan’s 1916 measurement of Planck’s constant, which has this anatomy and one simplification. A contact voltage between his photoemitting metal and his collector shifted every reading by the same amount, as $\mathcal{M}$ shifts every point of the Hubble diagram, and a slope carried the physics that a level could not touch.

Millikan wanted one number. Removing his nuisance parameter left that number identified, and five observations measured it to under one percent. The Friedmann model wants two, and removing $\mathcal{M}$ leaves them entangled with each other, so a second degeneracy survives the first. Section 5 is about that second degeneracy. It belongs to the geometry of the problem rather than to any instrument, and no supernova measurement removes it.

3 The Feature That Made It Possible

None of this helps while the noise is 0.4 magnitudes and the signal is 0.25.

Phillips [1993] found the fix. Plotting the rate at which a supernova fades after its peak against how bright that peak was, he found a tight relation: slower-declining supernovae are brighter. Writing Δm_{15} for the drop in B -band magnitude over the fifteen days after peak,

$$M_B^{\text{peak}} \approx M_B^{\text{ref}} + \alpha (\Delta m_{15} - 1.1), \quad \alpha \approx 0.75 \text{ mag}, \quad (5)$$

with α here the Hamuy et al. [1996] calibration. Correcting for it cuts the dispersion from 0.4 to about 0.15 magnitudes.

That factor of 2.7 in the noise is a factor of seven in the number of supernovae needed for a given precision. Against the open matter-only model the signal at $z \approx 0.5$ is 0.22 magnitudes, and detecting it at three standard errors takes about thirty supernovae at $\sigma = 0.4$ and about four at $\sigma = 0.15$. Both teams had roughly the number of events that the pre-Phillips dispersion would have required just to see the effect, with none left over to separate it from anything else.

The half of the discovery that came from data

The story so far credits theory. General relativity fixed the functional form, three parameters remained, and fifty supernovae sufficed where a theory-free learner would have needed thousands. Half the discovery does not fit that account.

Δm_{15} came out of data. No theory of Type Ia explosions predicted that decline rate tracks peak luminosity, and Phillips did not derive it. He plotted decline rate against absolute magnitude for nine nearby supernovae and fitted a line. The relation has since been partly rationalised—more radioactive nickel yields both a brighter peak and slower-cooling ejecta—and it has never been deduced. Thirty years on it remains an empirical regularity, recalibrated with every new survey. A theorist in 1993 holding the correct progenitor model could not have written (5), and a theorist today still cannot.

So the supernova discovery was a hybrid. A deductively specified hypothesis class was bolted to an inductively discovered feature, and neither half would have done. Without Friedmann there is no curve to fit and no meaning for the fitted numbers. Without Phillips the scatter is twice the signal and no amount of theory recovers it. The one step where a data-driven procedure did the indispensable work is the one step where no theory was available to do it instead.

The pattern has continued. Modern supernova cosmology runs on SALT2 and SALT3, light-curve models trained on data rather than derived from explosion physics, with training sets that grow every survey. Feature extraction in this field has become more machine-learned over time. The hypothesis class has stayed Friedmann’s.

4 The Signal

Table 1 gives the distance modulus predicted by three cosmologies.

Table 1: Predicted distance modulus for $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, by numerical integration. The last two columns give the excess faintness of flat Λ CDM over the open matter-only model and over Einstein–de Sitter.

z	Flat Λ CDM (0.3, 0.7)	Einstein–de Sitter (1.0, 0.0)	Open (0.3, 0.0)	Λ –open	Λ –EdS
0.10	38.32	38.21	38.25	+0.07	+0.11
0.25	40.50	40.27	40.36	+0.14	+0.24
0.50	42.26	41.86	42.05	+0.22	+0.40
0.75	43.33	42.82	43.08	+0.25	+0.51
1.00	44.10	43.50	43.84	+0.26	+0.60

At $z \approx 0.5$ the models differ by 0.22 to 0.40 magnitudes, against a per-supernova error of about 0.22 magnitudes. Each event carries roughly one standard deviation of signal. Both teams observed a weighted mean excess of about $+0.25 \pm 0.06$ magnitudes.

Naive stacking would put forty such events at $\sqrt{40} \approx 6$ standard errors. Both teams claimed about three. The gap is the subject of the next section, and it is the most interesting statistical feature of the episode.

The published estimates, all of which impose flatness, are in Table 2.

Table 2: Published estimates, all imposing $\Omega_M + \Omega_\Lambda = 1$. Statistical uncertainties only. The deceleration parameter is $q_0 = \Omega_M/2 - \Omega_\Lambda$; negative means accelerating.

Analysis	$\hat{\Omega}_M$	$\hat{\Omega}_\Lambda$	$\hat{q}_0$
Perlmutter et al. (SCP)	0.28 ± 0.09	0.72 ± 0.09	-0.58
Riess et al. (HZT), MLCS	0.24 ± 0.10	0.76 ± 0.10	-0.64
Riess et al. (HZT), template	0.20 ± 0.10	0.80 ± 0.10	-0.70
Planck 2018 (CMB + BAO)	0.315 ± 0.007	0.685 ± 0.007	-0.528

5 What the Supernovae Actually Measured

The Fisher information for (3) is

$$\mathcal{I}_{jk} = \sum_i \frac{1}{\sigma_i^2} \frac{\partial \mu(z_i)}{\partial \theta_j} \frac{\partial \mu(z_i)}{\partial \theta_k}, \quad (6)$$

and inverting it bounds the variance of any unbiased estimator. The calculation needs only the redshifts, the error bars, and the model. It does not need the magnitudes. Table 3 reports it for four designs, each with 18 nearby supernovae and $\mathcal{M}$ profiled out.

Table 3: Fisher standard errors for four observing designs, each paired with 18 low-redshift supernovae at $\sigma = 0.17$ mag and profiled over $\mathcal{M}$. High-redshift errors are 0.22 mag, except the last row, which uses 0.25.

High- z design	n	z range	SE(Ω_M)	SE(Ω_Λ)	SE($0.8\Omega_M - 0.6\Omega_\Lambda$)
SCP-like	42	0.18–0.83	0.44	0.65	0.10
Truncated	42	0.18–0.62	0.81	0.98	0.13
Bunched	42	0.35–0.55	1.30	1.50	0.12
HZT-like	16	0.16–0.62	1.32	1.56	0.19

The individual density parameters are not measured. The SCP-like design gives a standard error of 0.44 on Ω_M and 0.65 on Ω_Λ . The first exceeds the estimate itself, 0.28; the second is 90% of the estimate 0.72. Neither parameter is determined at all in the ordinary sense.

One combination is measured well. The forecast standard error on $0.8\Omega_M - 0.6\Omega_\Lambda$ is 0.10, and Perlmutter et al. [1999] report that combination as -0.2 ± 0.1 . A calculation using nothing but redshifts and error bars reproduces the published error bar. The observing design fixed the 1998 constraint; the details of the photometry did little to it.

The reason is a near-degeneracy in (1). Over the observed range, raising Ω_M and raising Ω_Λ bend $\mu(z)$ in nearly opposite ways, with $\partial\mu/\partial\Omega_M \approx -1.1 \partial\mu/\partial\Omega_\Lambda$ at $z = 0.5$ and a ratio nearer -1.5 when weighted across the whole sample. Raising Ω_M by one unit and Ω_Λ by about 1.5 therefore leaves the Hubble diagram nearly unchanged. Goobar and Perlmutter [1995] derived this geometry before the data existed and designed the survey around it.

Design beats sample size

The lower rows of Table 3 price the observing strategy. Keep the 42 supernovae and truncate the sample at $z = 0.62$ and the standard error on Ω_M nearly doubles. Bunch all 42 near $z = 0.45$ and it triples. The well-measured combination barely moves in either case, from 0.10 to 0.13 and 0.12.

Reach in redshift buys separation of the two parameters. It buys almost nothing on the combination. That is why the events beyond $z = 0.6$ were worth their cost in telescope time: they rotate the confidence ellipse, and rotation is what tells Ω_M from Ω_Λ . Where you put the observations mattered more than how many you took, which is the same lesson Millikan’s spectral range teaches in the photoelectric case.

Here the parallel with Millikan ends, and the difference is the harder half of the problem. Millikan’s equation had one slope, so removing the contact voltage left it fully identified. The Friedmann model has two shape parameters, and removing $\mathcal{M}$ leaves them tangled with each other. A second degeneracy survives the first. The contact voltage was an instrument nuisance that better instruments removed; this one is a property of the geometry, and no supernova measurement removes it.

Where the three sigma came from

The published significance therefore rests on more than the supernovae.

Measured by the marginal standard error on Ω_Λ alone, the SCP-like forecast puts $\Omega_\Lambda = 0.72$ just 1.1 standard errors from zero. The physical restriction $\Omega_M \geq 0$ does the rest. Setting $\Omega_\Lambda = 0$ and minimising over non-negative Ω_M raises χ^2 by 3.8 in the same forecast, two standard errors. No non-negative matter density reproduces the observed curvature with $\Omega_\Lambda = 0$, and that is what excludes a decelerating universe. The size of $\hat{\Omega}_\Lambda$ on its own does not.

Impose flatness instead and the picture changes. The same forecast gives a standard error of 0.070 on Ω_M , putting Einstein–de Sitter ten standard errors away. Flatness is what turns a two-parameter problem into a one-parameter problem, and it is what the published ± 0.09 describes.

Neither prior is controversial. The cosmic microwave background supplies flatness, and its constraint in the $(\Omega_M, \Omega_\Lambda)$ plane runs nearly orthogonal to the supernovae’s, so the two together determine what neither determines alone. Both priors are nonetheless external to the supernova data, and the published error bars do not survive their removal.

Partial identification of this kind will be familiar to economists. The data pin down a linear combination of structural parameters; the rest comes from a restriction imported from outside. The estimate is only as good as the restriction, and a careful presentation states the restriction alongside the number.

6 Two Teams, and the Objections

The two groups shared almost nothing but the answer. They used different telescopes and observing strategies. The High-Z team fitted multi-colour light-curve templates with a shape

parameter; the Berkeley team stretched a single-colour template along the time axis. Their K -corrections, which translate a redshifted spectrum back to a common rest-frame filter, came from different spectral templates. Their colour corrections used different estimators. Their high-redshift samples barely overlapped.

Their best-fit matter densities, 0.28 and 0.24, differ by less than half a standard error.

That agreement tests less than it appears to in one respect and more in another. Both numbers assume flatness, and under flatness the supernovae measure a single parameter, so two teams agreeing on one number with ± 0.10 error bars is a weak test. Against that, a systematic error in any pipeline step would have had to be reproduced by an independent implementation to survive. Replication across methods tests the standardisation. It does not test the statistics.

Both teams then had to answer three objections, all of which proposed that some systematic effect made distant supernovae look faint without any acceleration.

Grey dust. A screen of intergalactic dust that absorbs all wavelengths equally would dim distant supernovae without the reddening that normally betrays dust. Both teams checked the colour residuals. Dust of the required amount would leave a correlation between colour excess and magnitude offset. None was found at the level needed.

Luminosity evolution. If supernovae at $z \approx 0.5$ were intrinsically fainter than nearby ones, because their progenitors were younger or differently composed, the signal would be spurious. The teams compared the decline-rate and colour distributions of near and distant events, and compared their spectra. Within measurement error the distant supernovae looked like nearby ones of the same light-curve shape.

Selection. A magnitude-limited survey preferentially finds intrinsically bright events at high redshift, which biases the measured magnitudes the wrong way for acceleration but must still be quantified. Both teams simulated their own selection functions and found the effect small.

None of these checks is decisive on its own, and the teams said so. The decisive test was the one in the next section, and it took six more years.

7 The Test That Came Later

The 1998 fits estimated parameters from the data they fitted. The test described here estimated nothing.

Take the flat model with $\Omega_M = 0.28$ and $\Omega_\Lambda = 0.72$, fitted to supernovae below $z = 0.85$, and ask what it says about $z = 1.5$. Referred to Einstein–de Sitter it predicts an excess faintness of +0.41 magnitudes at $z = 0.5$ and only +0.74 at $z = 1.5$. The curve flattens, because dark energy dominates late and matter dominates early.

Now take the leading systematic explanation of the same 1998 data. Grey intergalactic dust and luminosity evolution both dim distant supernovae by amounts that grow with distance. Calibrated to reproduce the observed +0.41 at $z = 0.5$, a dimming proportional to z predicts +1.24 at $z = 1.5$ and keeps climbing.

Half a magnitude at $z = 1.5$ is two to three per-supernova standard deviations, which a handful of events settles. Referred to an empty universe instead, the prediction is sharper: Λ CDM puts

Table 4: Excess faintness relative to Einstein–de Sitter under the flat model fitted in 1998, and under a dimming proportional to z calibrated to match it at $z = 0.5$. Per-supernova errors at these redshifts were 0.2 to 0.3 magnitudes.

z	Λ CDM	Dimming $\propto z$	Difference
0.5	+0.41	+0.41	0.00
1.0	+0.62	+0.83	−0.20
1.5	+0.74	+1.24	−0.50

supernovae beyond $z \approx 1.4$ *brighter* than the benchmark, the signature of the deceleration that preceded the dark-energy era. No model in which distant supernovae are dimmed by dust or by evolution produces a change of sign.

[Riess et al. \[2004\]](#) collected the supernovae at $z > 1$ with the Hubble Space Telescope and found the turnover. The 1998 parameters had no freedom left to accommodate the result. That observation is the out-of-sample test of the model. The three sigma of 1998 measured parameters within it.

8 What They Could Not Settle

Type Ia supernovae measure one function, the luminosity distance $d_L(z)$. Every quantity above is a functional of that one curve. Two theories delivering the same $d_L(z)$ over the observed range are observationally identical to these data, whatever else they say.

Three classes of theory do so. A cosmological constant, with equation-of-state parameter $w = -1$, is one. A rolling scalar field with $w(z)$ within a few percent of -1 is a second; current data bound w only to about 0.03, giving $w = -1.03 \pm 0.03$. Modified theories of gravity that reproduce the same expansion history without any dark-energy fluid are a third. The 1998 data select the shape of $d_L(z)$. They do not select the physics generating it.

Distinguishing them needs a different observable. [Riess et al. \[2004\]](#) and the later surveys pursued one, the effect of w on $d_L(z)$ at higher redshift. The growth rate of large-scale structure supplies the other: modified gravity and dark energy predict different growth rates even when they produce identical expansion histories, so growth separates them where distances cannot. Theory names the discriminating observable. No one would measure growth rates to test dark energy without a theory predicting that the two differ.

And the mechanism remains unexplained. Quantum field theory predicts a vacuum energy density some 10^{120} times the observed Λ [[Weinberg, 1989](#)]. Riess, Schmidt and Perlmutter measured the curvature of the Hubble diagram to a few percent. The discrepancy between that measurement and the theory of what it measures is the largest in physics.

9 The Machine-Learning Reading

The vocabulary of statistical learning fits this episode closely, and the preceding note sets out why the translation is more than a conceit [[Sargent, 2025](#)]. Four things a trained network could

not deliver here mark the boundary of what fitting a curve accomplishes.

Start with what the physics supplies. Friedmann’s equations narrow the admissible curves from all functions $m_B(z)$ to a three-parameter family: $\mathcal{M}, \Omega_M, \Omega_\Lambda$. The restriction follows deductively from general relativity, the assumption of a homogeneous and isotropic universe, and the scaling of matter density with expansion. Each had been tested independently long before anyone looked for a high-redshift supernova.

Compare the alternative. The modest three-layer network priced in the preceding note carries 593 free parameters against the Friedmann model’s three, and wants some six thousand labelled examples where the Friedmann model wants four. Perlmutter had 42 supernovae. Riess had 50. The compilations of the 2010s, with several hundred, fall short by an order of magnitude, and today’s samples of a few thousand by a factor of three.

Give the network enough data and it would eventually trace the curve. Four things would still be beyond it.

It could not identify the learned function as $5 \log_{10} \tilde{d}_L(z)$ with $\tilde{d}_L$ satisfying a particular integral from general relativity. The curve would be a curve.

It could not extract Ω_M and Ω_Λ from its weights, because the weights do not decompose that way without the Friedmann equations imposed as an external constraint.

It could not relate the curvature to the deceleration parameter $q_0 = \Omega_M/2 - \Omega_\Lambda$, and so could not answer the question everyone wanted answered.

And it could not predict the $z > 1$ turnover of Section 7. Trained on data below $z = 1$, it would extrapolate the accelerating trend, because nothing in its architecture requires a transition. The observation that settled the 1998 result was a prediction of exactly the kind a function approximator does not make.

10 What This Says About AI in the Sciences

The preceding note argued that a correctly specified model can turn a problem needing thousands of observations into one needing five, and that no volume of data on the wrong observable substitutes for a theory naming the right one. The supernova episode bears both out. It also complicates them in a way the photoelectric case does not, and the complication is the more interesting half.

Induction did indispensable work here. Phillips found his relation by fitting curves to light curves; no theory predicted it, and none derives it now. Without it the scatter is twice the signal, and a perfectly specified Friedmann model fitted to unstandardised magnitudes recovers nothing. Without Friedmann the standardised magnitudes are a cloud of points with no parameter in them worth naming. The discovery required a deductively specified hypothesis class and an inductively discovered feature, and neither half was optional. Anyone arguing that theory beats learning, or that learning will eventually dispense with theory, has to explain this case away.

The pattern has persisted. SALT2 and SALT3, on which modern supernova cosmology runs, are light-curve models trained on data rather than derived from explosion physics, and their training sets grow with every survey. Feature extraction in this field has become more machine-learned

over time. The hypothesis class has stayed Friedmann’s. The division of labour is stable, and it does not fall along the line the usual arguments assume.

On the value of scale the supernova numbers are sharper than the photoelectric ones, because here the sample size can be held fixed while the design varies. Every row of Table 3 has the same 42 supernovae. Bunching them near $z = 0.45$ triples the standard error on Ω_M and leaves the well-measured combination almost untouched. Only the aim of the telescope changes across the rows, and the model is what aimed it.

The supernova case then adds a limit with no photoelectric counterpart. Millikan’s slope was identified; the density parameters are not. Fifty supernovae fix one combination of the two and leave the other open, so the published numbers close the gap with $\Omega_M \geq 0$, or with flatness taken from the microwave background. Neither restriction came out of the data being analysed. A procedure that returns a point estimate and a standard error while saying nothing about which restriction is carrying them has concealed the part of the inference a reader most needs to check. That failure is not peculiar to neural networks. Anyone who runs a regression and reports the output can commit it.

Section 8 states the outer limit. Unlimited supernova data would not separate a cosmological constant from a rolling scalar field, because the two imply the same $d_L(z)$. The separation waits on measurements of the growth of structure, which exist as a target only because a theory predicts that the two differ there.

11 Summary

The discovery of cosmic acceleration is a three-parameter weighted nonlinear regression whose functional form comes from general relativity. The regression is straightforward. Two other things were not. Standardising the supernovae by an empirical relation nobody had derived cut the noise by a factor of 2.7. Assembling supernovae at $z = 0.3$ to 0.8 put observations where the competing models differ by two or three tenths of a magnitude. Neither task is statistical.

Under flatness, the estimates of both groups give

$$\hat{q}_0 = \frac{\hat{\Omega}_M}{2} - \hat{\Omega}_\Lambda = -0.58, \quad \hat{\Omega}_\Lambda = 0.72 \pm 0.09, \quad z_t \approx 0.73,$$

where z_t is the redshift at which deceleration gave way to acceleration. Without flatness the individual parameters are not determined, and the inference to acceleration runs through the restriction $\Omega_M \geq 0$.

Fifty supernovae sufficed because the physics had fixed the functional form and left three parameters. The fifty could not separate two of those three, and the published numbers close that gap with a restriction the supernovae do not supply. Both facts belong in any account of what the data established.

The Fisher forecasts use the observed redshift distributions and quoted uncertainties, not the individual magnitudes; the parameter estimates are the two teams’ own published values. The significance figures attributed

to Riess et al. [1998], the calibration in (5), and the bound on w come from secondary sources and should be checked against the originals.

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